

Enhancing Customer Service Through LLM-Powered Email Drafting: An E.ON Data and Analytics Use Case

Aslihan Berber, Mohammed Yazan Aljleilati

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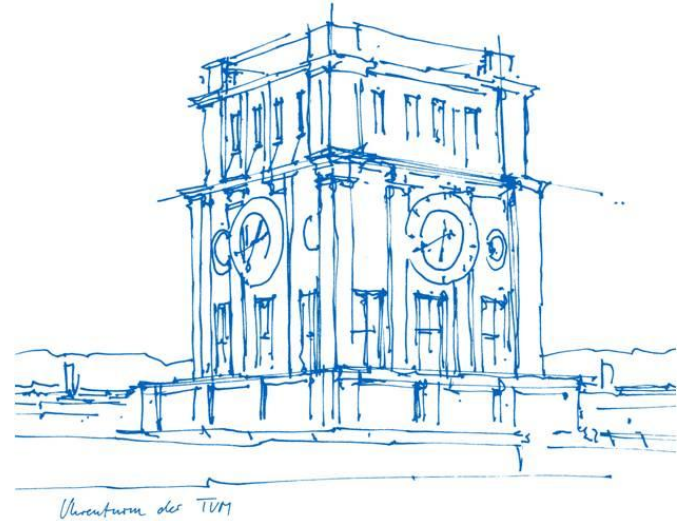
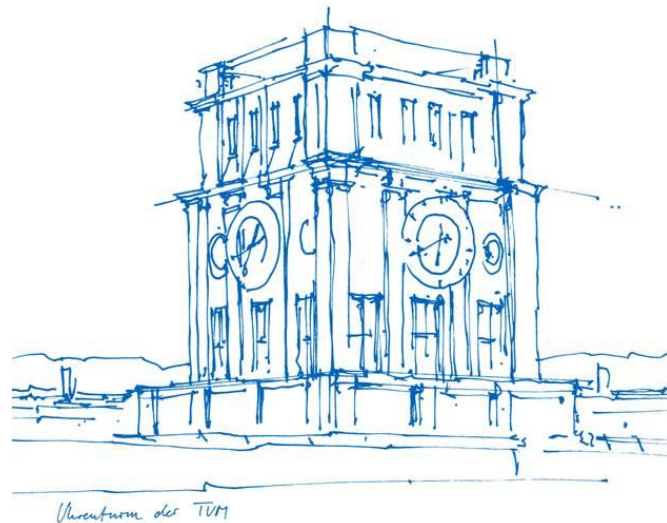


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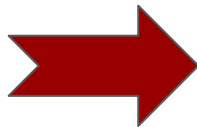


Context & Problem

- High volume of customer emails at E.ON
- Repetitive topics: billing, contracts, meter readings
- Email handling is a major operational overhead
(Thiergart et al., 2021)

Why LLMs

- LLMs enable instruction-following & fluent text generation
(Cai et al., 2025)
- Proven effective as drafting assistants in customer service
(Thiergart et al., 2021; Wulf & Meierhofer, 2023)
- Successful real-world deployments in knowledge work
(Ünlü, 2023)



Our Approach

- End-to-end, human-in-the-loop drafting system
(BCG, 2023; Costa & Ghosh, 2025)
- Created synthetic response pairs to input data
- Formal & friendly draft replies for agent support;
used TinyLama; a **compact open-source LLM** under realistic constraints
- Combined topic detection, summarization, and response generation
- Generated formal & friendly draft replies for customer agents

- 161 anonymized customer emails
- Personal information was removed or replaced
- Each Email-file contains one complete customer inquiry

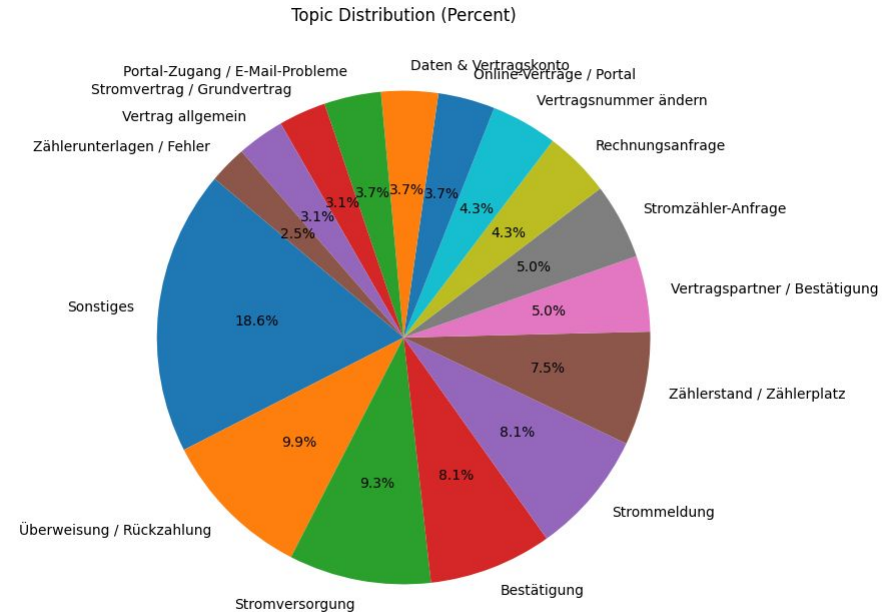
- Converted to JSON Lines
- No heavy preprocessing

```
{  
  "email_id": "134",  
  "raw_text": "Sehr geehrte Damen und Herren, wie in o.g.  
Telefonat besprochen ..."  
}  
{  
  "email_id": "24",  
  "raw_text": "Sehr geehrte Damen und Herren, bitte bereiten Sie  
mir ein Angebot für ..."  
}
```

Topic Detection, Clustering & Labeling



- BERTopic for semantic clustering
- 15 Clusters in total
- Each email assigned to a topic & confidence interval
- Manual refinement of topic labels
- Main topics: payments, electricity supply, meter readings, contracts



email_id	topic_id	topic_id	topic_label	topic_confidence
0	22	9	Online-Verträge / Portal	0.193802
1	103	10	Daten & Vertragskonto	1.000000
2	12	1	Stromversorgung	0.610613



Why a Small Model & Design Motivation

- Realistic compute constraints (Google Colab)
- No access to large proprietary models
- Exploratory focus on *practical feasibility*, not raw scale
- Interest in compact models for enterprise use



Chosen Model

- TinyLlama (1.1B parameters)
- Chat-aligned, instruction-following
- Built on a modern transformer architecture and has been pre-trained to follow natural-language instructions and generate assistant-responses
- Open-source & lightweight
- Libraries: PyTorch, Transformers, Peft, bitsandbytes, trl

Implementation: Fine Tuning Strategy (QLoRA)

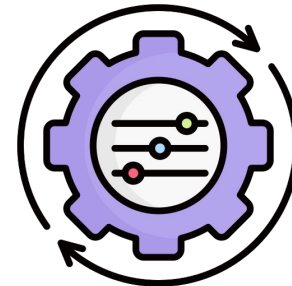


Training Approach

- Supervised instruction fine-tuning using a parameter-efficient optimization strategy
- Synthetic instruction-response pairs
- Two outputs per email: **formal & friendly**

Parameter-Efficient Optimization

- LoRA adapters introduces a small number of trainable parameters into selected transformer layers while keeping the base model frozen
- 4-bit quantization
- **QLoRA** for low-memory fine-tuning



e-on

- **Explicit role definition: E.ON customer service assistant**
- **Tone specified directly in the instruction**
- **Same input → different stylistic targets**

FORMAL:

"Du bist ein KI-Assistent, der Mitarbeitende im E.ON-Kundenservice unterstützt.
Formuliere eine formelle, klare und höfliche Antwort-E-Mail auf Deutsch."

FRIENDLY:

"Du bist ein KI-Assistent, der Mitarbeitende im E.ON-Kundenservice unterstützt.
Formuliere eine freundliche und weniger formelle Antwort-E-Mail auf Deutsch."

Implementation: Training Setup & Evaluation



Training Setup

- 80 / 20 train-validation split
- 600 update steps
- Gradient accumulation for stability

A validation loss of ≈ 1.30 means:

- The model assigns relatively high probability to the correct next tokens
- The generated outputs are **coherent and well-conditioned**
- The model is **not guessing randomly**
- The model is **not memorizing training data**

Evaluation Metrics	Value
Total Samples	322
Train/Validation split	257/65
Training Steps	600
Learning Rate	2×10^{-5}
Effective Batch Size	8
Training Loss	1.48
Evaluation/Validation Loss	1.30

Validation loss < training loss $\longrightarrow$ The model generalized instead of memorizing





Inference Pipeline

- Topic assignment using BERTopic
- Rule-based short summary generation
- Dual-tone response drafting (formal & friendly)

Controlled Decoding

- Deterministic decoding (beam search, 3 beams) -> so that the same email always produces the same output
- Repetition control to avoid redundant phrasing
- Reproducible and conservative outputs

Safety & Compliance Controls

- Explicit instruction- and rule-based prompting
- No hallucinated numbers, dates, or contact details
- Ask follow-up questions if information is missing
- Human-in-the-loop review remains mandatory

- **No Fact Invention**

If the customer email does not contain a contract number or amount, the model is explicitly instructed not to invent one.

- **Exact Reuse of Numbers**

If the customer writes "103,22 €", the model must reuse "103,22 €" exactly and is prohibited from rounding or modifying it.

- **No Assumed Actions**

The model is forbidden from stating that a payment was completed unless the customer explicitly confirms it in the email.

- **Missing Information Handling**

When required details are missing, the model must ask 2-4 concrete follow-up questions instead of guessing.

- **No Invented Dates or Timelines**

Dates such as "am 22.02.2022" or "seit 2019" may only appear if they occur verbatim in the customer email.

- **Deterministic Output**

The same customer email always produces the same response due to deterministic decoding without sampling.

- **Signature Enforcement**

If the model repeats or omits the closing, the system removes duplicates and enforces a single E.ON signature.



Validation: Lexical Diversity Check



Objective:

Detect whether the model collapses into repetitive, template-like phrasing

Method:

- Lexical entropy over generated responses
- Measures **vocabulary diversity**
- Penalizes repeated or formulaic wording

Input data (unseen emails, n=10):

All generated outputs (**formal + friendly combined**)

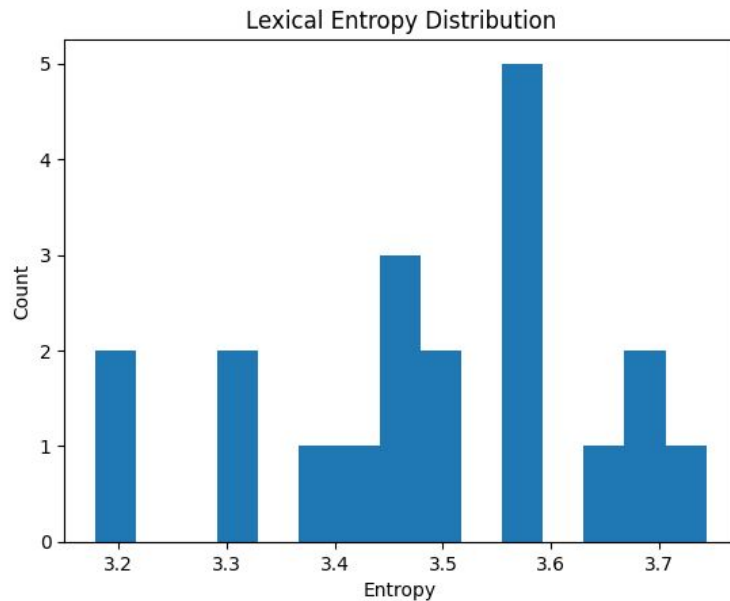
Text normalized (lowercasing, whitespace cleanup)

Implementation details:

- Tokenize responses into word tokens
- Compute token frequency distribution
- Calculate entropy

Results:

- Mean entropy = **3.49** -> **good diversity for emailing**
- Standard deviation = **0.15** -> **stable behavior**
- No low-entropy collapse observed



→ Responses showed **controlled linguistic variation**, not copy-paste templates

Validation: Semantic Variety Check



Objective:

Ensure responses vary in phrasing **without drifting in meaning**

Problem addressed:

Lexical variation alone can hide semantic inconsistency

Embedding model used:

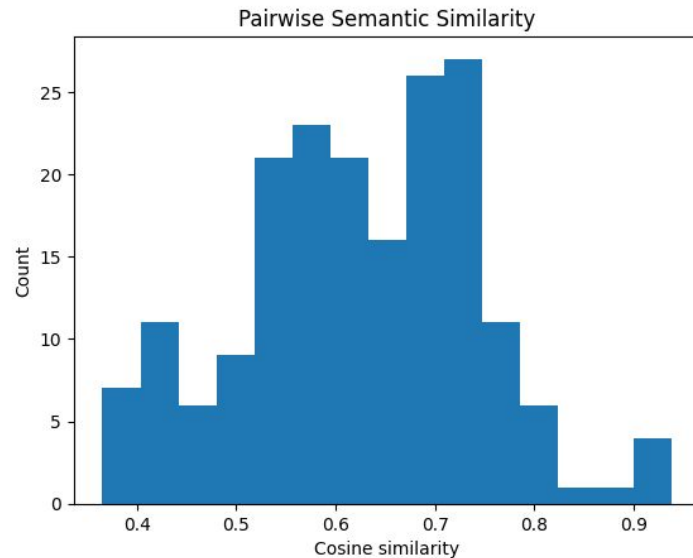
- `all-MiniLM-L6-v2` (Sentence-Transformers)
- Lightweight, fast, suitable for short texts

Implementation details:

- Encode each generated response into sentence embeddings
- Compute **pairwise cosine similarity**
- Exclude self-similarity (diagonal removed)

Results:

- Mean cosine similarity = **0.62**
- Standard deviation = **0.11**
- Majority of pairs fall in mid-to-high similarity range



- Responses remain **semantically aligned**
- Still, allows phrasing and structural variation

Hallucination Check: Factual Grounding Validation



Why hallucination check matters

- Incorrect facts can cause billing errors, legal risk, and GDPR violations
- Especially critical in regulated domains like energy

What we validate

- New facts introduced by the model vs. customer email
- Checked entity types:
 - Numbers (amounts, contract IDs)
 - Dates
 - Email addresses
 - URLs
 - Phone numbers

How it works

- Rule-based extraction using regex
- Normalization to handle formatting differences (e.g., “232 671” → “232671”)
- Per-response hallucination flagging/risk detection

(Set difference: response facts – input facts) =

Any new entity ⇒ hallucination flag

Final results (unseen emails, n = 10)

- Formal responses: **0% hallucination**
- Friendly responses: **0% hallucination**
- No new factual entities introduced

→ Model reliably grounds responses in customer input

→ Suitable for internal drafting with human review



Model

Merged model: /content/drive/My
Drive/DAIA_EON/tinyllama_eon_merged

E.ON Customer Email Assistant

Paste customer email here:

Paste the customer email...



Generate Formal

Generate Friendly

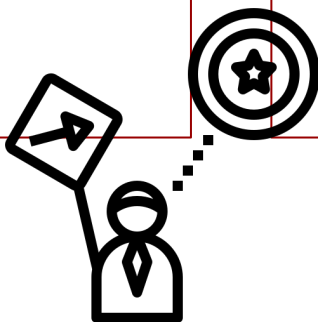
Paste a customer email above to begin.

Key Limitations

- Compact **TinyLlama model** (performance ceiling)
- **Synthetic training data** (no real agent replies)
- No **human-in-the-loop user study**
- Limited evaluation sample size
- Limited data size for training
- No **backend or RAG integration**

Future Outlook

- Testing **larger models** under same constraints
- Testing with more data
- Incorporating **agent feedback / edits**
- Running **human-in-the-loop evaluations**
- Integrating **retrieval-augmented generation** to provide accurate and factually grounded answers
- Including a more personalized tool and user interface



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Thank You!

Munich, 30. January 2026

